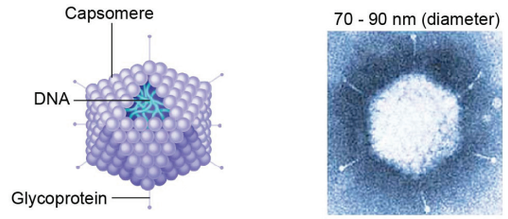
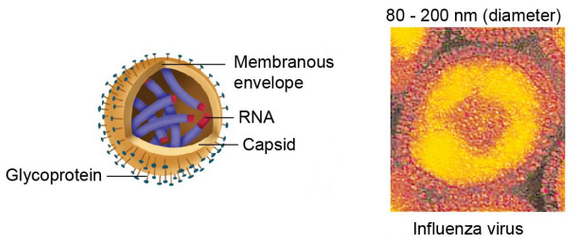
Animal virus: Virus that infect animals. e.g. Adenovirus, Retrovirus(HIV), Influenza virus, Polio virus.



Adenovirus



Influenza virus

Figure 22.4 Animal virus

Bacteriophages: Virus that infect bacterial cells. e.g. T4 bacteriophage.

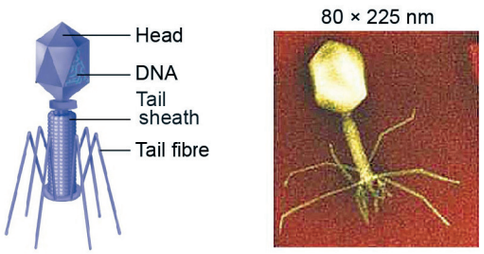


Figure 22.5 T4 bacteriophage

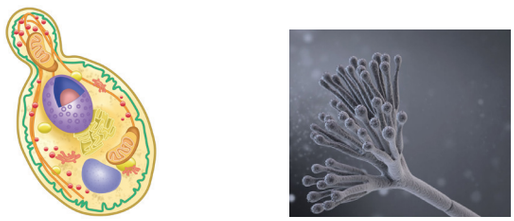
22.1.3 Fungi

They lack chlorophyll, hence depend on living or dead host for their nutritional needs. Fungi living on living hosts are called parasites, and those living on dead organic matter are called saprophytes. The body of the fungus is called **thallus**.

Single celled yeast ranges from 1 to 5 µm in width. They are spherical in shape. Flagella are absent and hence they are non-motile. In the case of multicellular forms, thallus is called mycelium. **Mycelium** is a complex of several thin filaments called **hyphae** (singular: Hypha).

Each hypha is 5 to 10 µm wide. They are tube like structures filled with protoplasm and cellular organelles. Cell wall is made up of cellulose or chitin. Cytoplasm contains small vacuoles filled with cell sap, nucleus, mitochondria, golgi body, ribosomes, and endoplasmic reticulum. Food material is stored in the form of glycogen or oil globules.

They reproduce vegetatively (binary fission, budding and fragmentation), asexually (spore formation-conidia) and sexually (male and female gametangium are called antheridium and oogonium).



Yeast cell *Penicillium*

Figure 22.6 Structure of fungi

22.1.4 Prions

The term 'prion' was coined by Stanley B. Prusiner in 1982. Prions are **viral particles** which contain only proteins. They do not contain nucleic acid. They are infectious and smaller than viruses. Prions are found in neurons and are rod shaped. Prions induce changes in normal proteins. This results in the degeneration of nervous tissue.

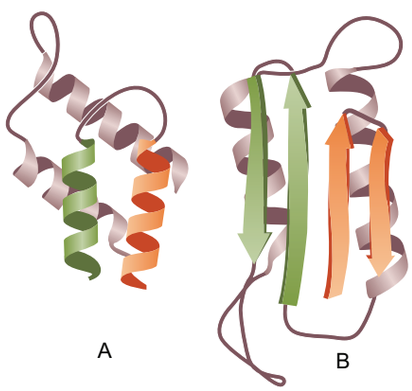


Figure 22.7 Normal (A) and Abnormal (B) prion protein

IX_SCI_EM_Unit-22_BIO.indd 260 12/7/2022 5:31:30 PM